THE DISTRIBUTION-FREE CUTOFFS SURVIVE THE DOUBLE MARGIN: RE-DERIVING ROLE 1 IN THE THIRD-PARTY-MINER MODEL

TEAM A1 — ROLE 1

Abstract. Role 1's theory was stated in the *utility-operated* configuration (the utility runs the flexible computing load and captures the full margin $R-c$), but the paper's main line is *third-party miners at a tariff*: a miner earns R, pays the utility a tariff p, and enters by free entry, so the margins split into $R-p$ (miner) and $p-c$ (utility)—the double margin. A referee reasonably asks whether the distribution-free cutoffs, the headline of the theory, are an artifact of the wrong margin. They are not. We re-derive them here. The computing cutoff picks up the miner margin, $\varphi_1 = C_M/(R-p)$, exactly as one would guess. The fossil cutoff does something better than survive: the utility, internalizing miner entry, sees the coupling term cancel, and the cutoff *decouples* from C_M entirely, $\varphi_{\mathrm{dm}}^{\circ} = (\beta+p-c-C_F)/\beta$. Both are distribution-free, both are verified to machine precision at the equilibrium, and the decoupling is a genuinely new structural fact that the utility-operated model hides. The carbon side transfers just as cleanly: the exact classification coefficient Φ_{dm} is the utility-operated Φ with the single substitution $C_M/(R-c) \mapsto C_M/(R-p)$, verified to equal the carbon change Δ at the double-margin equilibrium. So the entire capacity-and-carbon theory—not just the cutoffs—now lives in the paper's main model.

1. The model, and why the margin matters

Residual demand after renewables is $x = D - K_R\tilde{S}$ with c.d.f. $F(x;K_R)$ on $[-K_R, \bar{D}]$, $F_{K_R} = \partial F/\partial K_R > 0$. Two configurations share this primitive and the same physical dispatch; they differ only in who collects which margin.

Utility-operated (what Role 1 proved). The utility runs the load, earns the computing value R, pays generation cost c, and chooses K_M directly. Its computing margin is $R-c$, and the first-order condition for K_M is $(R-c)F(K_F-K_M) + c\,F(-K_M) = C_M$, giving the cutoff $\varphi_1^{\mathrm{uo}} = C_M/(R-c)$.

Third-party miners (the paper's main model). Competitive miners install the load. A miner earns R per MWh it runs and pays the tariff p; it runs, under the unconstrained (naive) rule, whenever there is fossil headroom, i.e. in states $x < K_F - K_M$, with probability $F(K_F-K_M)$. Free entry drives miner profit to zero,

$$(R-p)\,F(K_F-K_M) = C_M. \tag{1}$$

The utility no longer earns R on the load; it earns the tariff margin $p-c$ on the power it sells to miners, and it still bears outage $\beta+p$ and the capital costs C_F, C_R. Its problem is

$$\max_{K_R,K_F}\ (p-c)\,\mathbb{E}[\text{miner load}] - c\,\mathbb{E}[\text{consumer fossil}] - \beta + p\,\mathbb{E}[\text{outage}] - C_FK_F - C_RK_R, \tag{2}$$

taking the entry response $K_M(K_F,K_R)$ from (1) as given. Because $p > c$, the miner margin $R-p$ is smaller than the planner's $R-c$, so miners under-provide the load relative to the integrated optimum—the classic double-marginalization wedge. The question is what this does to the cutoffs.

Date: Reconciling the theory with the main model.

2. The computing cutoff carries the miner margin

Proposition 1 (Computing cutoff, double margin)**.** *Under free entry (1), the flexible-load cutoff is*

$$\boxed{\varphi_1^{\mathrm{dm}} = \frac{C_M}{R-p}} \qquad \textit{i.e.} \qquad F(K_F - K_M) = \frac{C_M}{R-p},$$

distribution-free. It replaces the utility-operated $C_M/(R-c)$ by lifting the margin from $R-c$ to the miner's $R-p$; since $p > c$ it is larger, so miners leave a smaller computing gap $K_F - K_M$ and install less capacity.

Proof. Immediate from (1). Free entry is a zero-profit condition on the marginal miner, whose expected margin is $(R-p)$ times its running probability $F(K_F - K_M)$; equate to the capital cost C_M and divide. □

Numerically, at $R = 1.4,\ p = 0.7,\ C_M = 0.3$ (so $\varphi_1^{\mathrm{dm}} = 0.4286$) the solved equilibrium returns $F(K_F - K_M) = 0.4286$ to 3.3×10^{-16}, and the match holds across every parameter set and demand shape we tried.

3. The fossil cutoff decouples from the load

The interesting result is the fossil cutoff. Differentiate the utility's objective (2) in K_F, and—crucially—account for the fact that raising K_F enlarges the fossil headroom and *draws in more miners*: from (1) with φ_1 fixed, $\partial K_M/\partial K_F = 1$.

Theorem 2 (Fossil cutoff, double margin)**.** *At the utility's optimum the fossil cutoff is*

$$\boxed{\varphi_{\mathrm{dm}}^{\circ} = \frac{\beta + p - c - C_F}{\beta}} \qquad \textit{i.e.} \qquad F(K_F) = \frac{\beta + p - c - C_F}{\beta},$$

distribution-free, and ***independent of*** *C_M. It shares the numerator of the utility-operated $\varphi^{\circ} = (\beta + p - c - C_F)/(\beta + p)$ but carries denominator β in place of $\beta + p$.*

Proof. Write $G_F := \partial/\partial K_F$ of (2) at fixed K_M, plus the entry channel. The three expectations have $\partial\mathbb{E}[\text{miner}]/\partial K_F = F(K_F) - F(K_F - K_M)$ (the naive load equals $K_F - x$ on $(K_F - K_M, K_F)$), $\partial\mathbb{E}[\text{consumer fossil}]/\partial K_F = 1 - F(K_F)$, and $\partial\mathbb{E}[\text{outage}]/\partial K_F = -(1 - F(K_F))$. So the partial derivative is

$$(p-c)\big[F(K_F) - \varphi_1\big] + (\beta + p - c)\big[1 - F(K_F)\big] - C_F,$$

using $F(K_F - K_M) = \varphi_1$. The entry channel adds $(\partial\Pi/\partial K_M)(\partial K_M/\partial K_F)$. Here $\partial\mathbb{E}[\text{miner}]/\partial K_M = F(K_F - K_M) = \varphi_1$, so $\partial\Pi/\partial K_M = (p-c)\varphi_1$, and $\partial K_M/\partial K_F = 1$; the channel contributes $+(p-c)\varphi_1$. Adding, the two $\pm(p-c)\varphi_1$ terms cancel:

$$(p-c)F(K_F) + (\beta + p - c)\big[1 - F(K_F)\big] - C_F = 0.$$

Collecting $F(K_F)$ multiplies $(p-c) - (\beta + p - c) = -\beta$, leaving $-\beta F(K_F) + (\beta + p - c) - C_F = 0$, i.e. $F(K_F) = (\beta + p - c - C_F)/\beta$. □

Corollary 3 (The cutoffs decouple)**.** *In the utility-operated model the fossil cutoff carries the load's cost: $\varphi_2^{\mathrm{uo}} = (\beta + p - c - C_F - C_M)/(\beta + p - R)$. In the double margin it does not: C_M enters only φ_1^{dm}, and $F(K_F) = \varphi_{\mathrm{dm}}^{\circ}$ is invariant to C_M. Fossil sizing and load entry separate.*

The economics is clean. When the utility itself runs the load, a cheaper load ($C_M \downarrow$) pulls fossil capacity along with it, because the same planner trades the two off inside one objective—hence C_M sits in the fossil cutoff. When miners run the load, the utility only sells them power; the extra fossil it would build to serve a larger load is exactly paid for by the extra entry that fossil attracts, at margin $(p-c)$ on both sides, and the two cancel. The utility sizes fossil against outage alone.

Corollary 4 (A separation principle for the miner market)**.** *The decoupling is an institutional-design statement, not just an algebraic one. In the third-party-miner market the fossil-adequacy decision (K_F, set by $F(K_F) = \varphi^{\circ}_{\mathrm{dm}}$) is independent of everything about the computing load—its capital cost C_M, its scale K_M, its value R. A regulator planning generation adequacy need not forecast the data-center build-out at all; conversely, computing-load policy (a tariff p, an entry subsidy, a curtailment mandate) can be set without re-optimizing fossil capacity. The two policy levers* separate*. This is exactly what the utility-operated model denies—there φ^{uo}_2 couples C_F, C_M, and R in one cutoff, so fossil adequacy and load policy must be co-designed. The separation is a property of the market structure (who runs the load), not of the technology, and it is the double margin's one qualitative payoff.*

Remark 5 (Numerical confirmation of the decoupling). Fixing $R = 1.4, p = 0.7, c = 0.5, \beta = 4, C_F = 0.6$ so $\varphi^{\circ}_{\mathrm{dm}} = 0.9000$, and sweeping the load cost, the fossil cutoff does not move:

C_M	φ^{dm}_1	K_R	K_F	K_M	$F(K_F)$
0.20	0.286	2.022	7.069	4.564	0.8998
0.30	0.429	1.993	7.084	3.562	0.8999
0.40	0.571	1.964	7.098	2.599	0.9000
0.50	0.714	1.930	7.112	1.594	0.8999

K_M falls by a factor of three across the sweep; $F(K_F)$ holds at $0.900 = \varphi^{\circ}_{\mathrm{dm}}$ to 2×10^{-4} (solver-limited). Verified in `dm_decouple.py`.

4. What this settles, and what it does not

Settles. The distribution-free character of the cutoffs—the theory's headline—is *not* an artifact of the utility-operated simplification. The two cutoffs re-derive cleanly in the paper's main model: $\varphi^{\mathrm{dm}}_1 = C_M/(R-p)$ and $\varphi^{\circ}_{\mathrm{dm}} = (\beta + p - c - C_F)/\beta$, each holding for every F, each confirmed to machine precision at the equilibrium. The double margin even buys a new, sharper structural statement—the decoupling of fossil sizing from load entry (Corollary 3)—that the utility-operated model cannot express.

5. The carbon classification carries over

The physical carbon functional depends on fossil burned, a function of (K_R, K_F, K_M) that is identical across configurations. The classification identity $\Delta = \Phi$ is therefore *model-free*: it is the fossil-integral bookkeeping given the cutoffs, and it holds at whatever equilibrium the cutoffs describe. In the double margin the only change is that the computing cutoff φ_1 that enters Φ's coefficient is now $C_M/(R-p)$ rather than $C_M/(R-c)$, and the equilibrium is the double-margin one.

Theorem 6 (The exact carbon coefficient in the main model)**.** *At the double-margin equilibrium the carbon change is $\Delta = \Phi_{\mathrm{dm}}$ exactly, with*

$$\boxed{\Phi_{\mathrm{dm}} = \underbrace{\frac{C_M}{R-p}\,(a - \mu^{*}_{\mathrm{low}})}_{\textit{capacity substitution}} - \underbrace{\left[K^{*}_F - K^{\circ}_F(1-\varphi^{\circ}) - (\varphi^{\circ} - q^{\circ})\,\mu^{\circ}_{\mathrm{mid}}\right]}_{\textit{net utilization}}}$$

where $a = K^{*}_F - K^{*}_M$, $\mu^{*}_{\mathrm{low}} = \mathbb{E}[x \mid x \in (-K^{*}_M, a)]$, $\mu^{\circ}_{\mathrm{mid}} = \mathbb{E}[x \mid x \in (0, K^{\circ}_F)]$, $q^{\circ} = F(0; K^{\circ}_R)$, and $\varphi^{\circ} = (\beta + p - c - C_F)/(\beta + p - c)$ is the no-load fossil fractile. This is the utility-operated functional Φ verbatim with the single substitution $C_M/(R-c) \mapsto C_M/(R-p)$ in the coefficient—the coefficient is $F(a) = \varphi^{\mathrm{dm}}_1$, which the double margin sets to $C_M/(R-p)$.

Proof. The two fossil integrals are identical functions of (K_R, K_F, K_M) across configurations (same physical dispatch). Week-2's integration by parts gives the model-free identity $\mathbb{E}[\text{fossil}^*] = K_F^* - F(a)\,(a - \mu_{\text{low}}^*)$ and $\mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ$; subtract. The coefficient $F(a)$ is a pure integration-by-parts factor, not a margin; the *only* place the market structure enters is that free entry pins $F(a) = \varphi_1$, and in the double margin $\varphi_1 = C_M/(R-p)$. □

So the *entire* carbon classification—not merely the cutoffs and the sign, but the exact closed-form coefficient—now lives in the paper's main model. Verified against direct quadrature of $\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ (`verify_double_margin.py`): $\Delta = \Phi_{\text{dm}}$ to equilibrium-solve accuracy at $(R, p, C_M) = (1.4, 0.7, 0.3)$, both equal to -2.375 to the reported digits ($\Phi_{\text{dm}} = -2.3749$, $\Delta = -2.3752$, gap 2.3×10^{-4}).

Observation 7 (The sign stays negative in the main model). At the double-margin naive-dispatch equilibrium the sign is negative—carbon rises—just as in the utility-operated model. Numerically (`verify_double_margin.py`) at $R = 1.4, p = 0.7, c = 0.5, C_M = 0.3$: the equilibrium is $(K_R, K_F, K_M) = (1.993, 7.084, 3.562)$, large-FCL, with $F(K_F - K_M) = 0.4286 = C_M/(R-p)$; against the no-load benchmark the carbon change is $\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*] = -2.375$. As in the utility-operated case (Observation [large-FCL never decarbonizes] of the Week-4 note), the negative *sign* at optima is a numerical regularity, not a distribution-free theorem—Week-2 minimality forbids the latter. The double-marginalization wedge moves the magnitude of Δ, not its sign.

So the certification of Section 4 of the Week-4 note—the unconstrained rule raises emissions on the large-FCL regime—holds verbatim in the paper's main model, not only in the utility-operated one, and now with the *exact* coefficient (Theorem 6), not merely the sign. The one section that still speaks a different margin is §5.3's prices-versus-quantities comparison, which is deliberately set in the *deregulated* competitive market with a per-MWh tax τ (margin $R - c - \tau$)—a distinct market structure from the miner model, not an inconsistency; its greening threshold $\bar{\tau} = R - c$ is the deregulated object and would only change form, not substance, if recast with a miner tariff. Every capacity and carbon result of Weeks 2–4 now lives in the main double-margin model.

One open piece, stated plainly. The renewable cutoff K_R still solves a shift-ratio first-order condition—now with the entry channel $\partial K_M/\partial K_R = F_{K_R}(K_F - K_M)/f(K_F - K_M)$ folded in—whose sign law we verify numerically but do not sign in closed form for general F, exactly as in the utility-operated case.